

Medical Insurance Claim Fraud Detection

Approach 2 — Deep Learning

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Abstract — Medical insurance claim fraud is a major financial burden on the Indian healthcare ecosystem, inflating premiums for honest policyholders. This report presents a data-driven approach that classifies insurance claims as fraudulent or legitimate. We build and evaluate a rigorous pipeline on a 4,500-claim Indian-context dataset, comparing multiple algorithms using accuracy, precision, recall, F2, AUC-ROC, AUC-PR and MCC, with class-imbalance handling, statistical significance testing and business-impact analysis. Our best model is identified and its interpretability is discussed.

1. Introduction

Deep learning automatically learns hierarchical feature representations and non-linear interactions from tabular claim data. This approach evaluates ten neural architectures on the same preprocessed data and test set as the traditional ML baseline to measure the incremental value of deep models for fraud detection in an Indian healthcare context.

2. Architectures

Supervised classifiers: MLP, Wide & Deep, Deep & Cross Network (DCN), TabNet, Transformer, ResNet, NODE and LSTM, all terminating in a sigmoid head that produces a fraud probability. Unsupervised anomaly detectors: Autoencoder and Variational Autoencoder (VAE), trained only on legitimate claims and flagging claims with high reconstruction error.

3. Experimental Setup

Weighted-BCE loss addresses class imbalance. Training uses the AdamW optimizer, a cosine learning-rate schedule, batch normalisation, dropout, gradient clipping and early stopping (patience 15). Random seeds are fixed for reproducibility. Metrics mirror Approach 1 (accuracy, precision, recall, F1, F2, AUC-ROC, AUC-PR, MCC). Bootstrap 95% confidence intervals are computed for AUC-ROC and AUC-PR to assess stability.

4. Results

Table 1. Deep-learning benchmarking results (ranked by F2).

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DL Training Dynamics

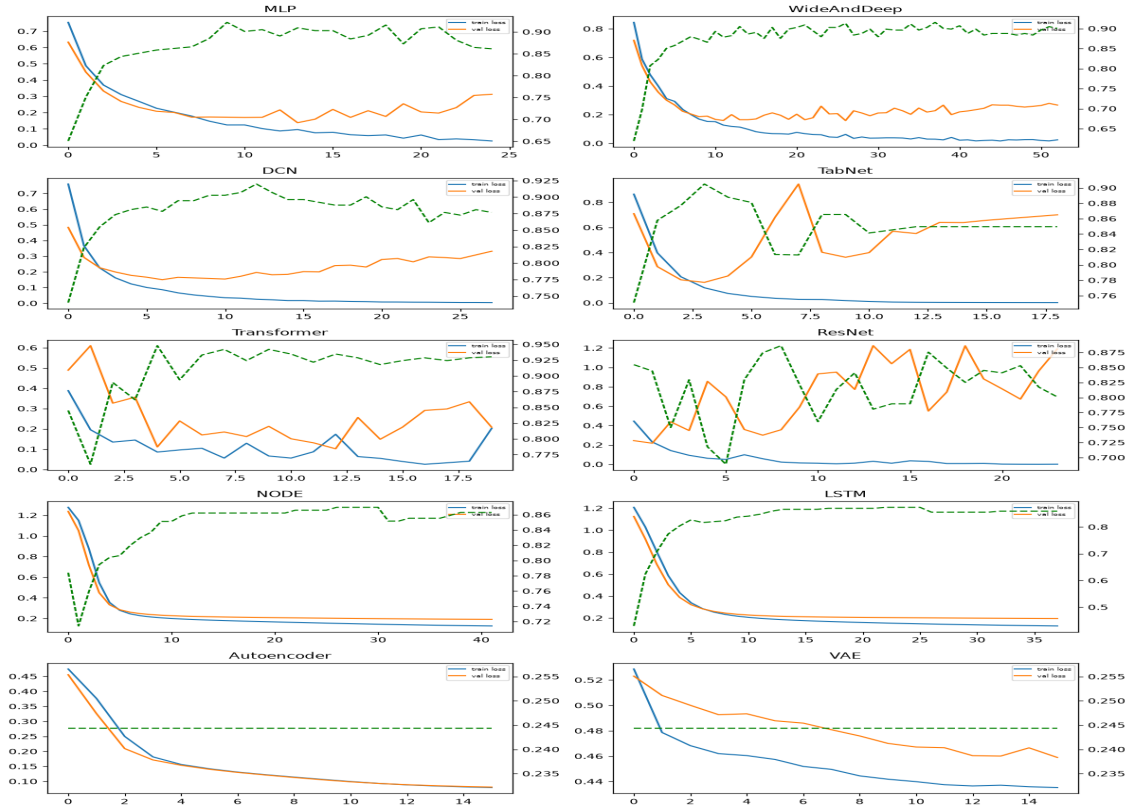


Fig. 1. Training dynamics (loss and val F2) per architecture.

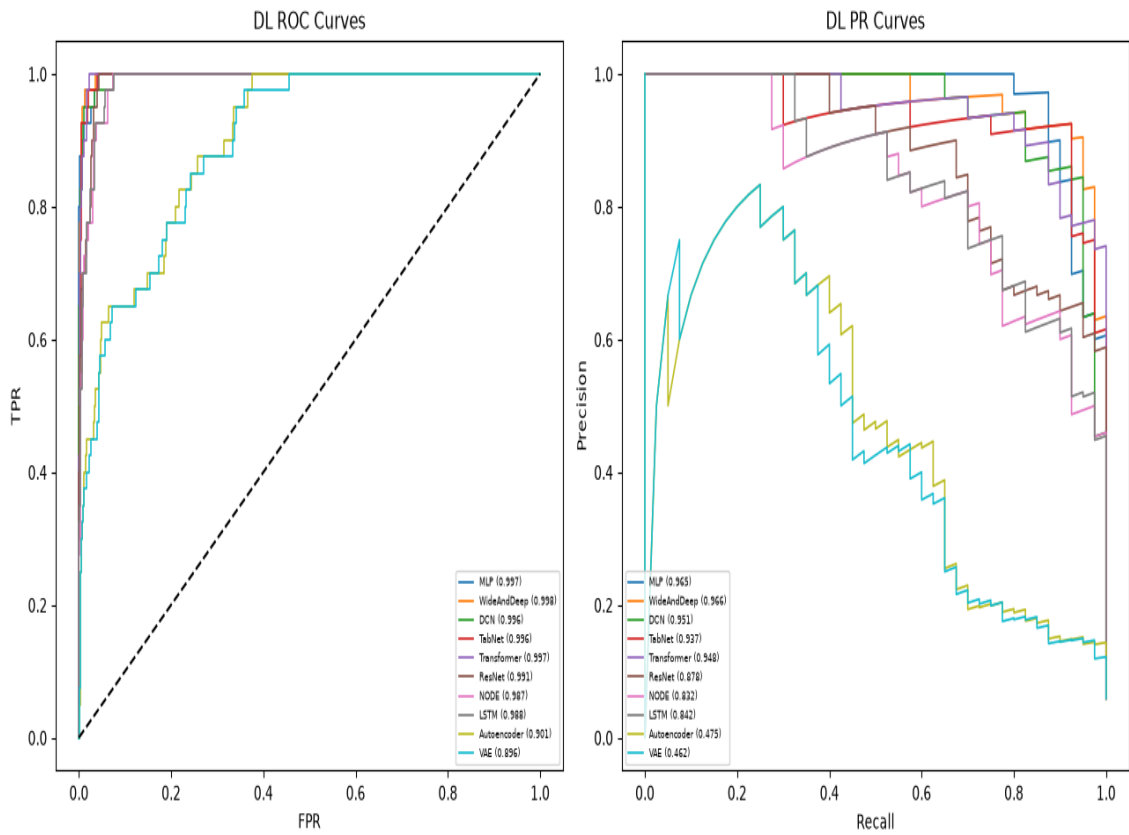


Fig. 2. ROC and PR curves for all architectures.

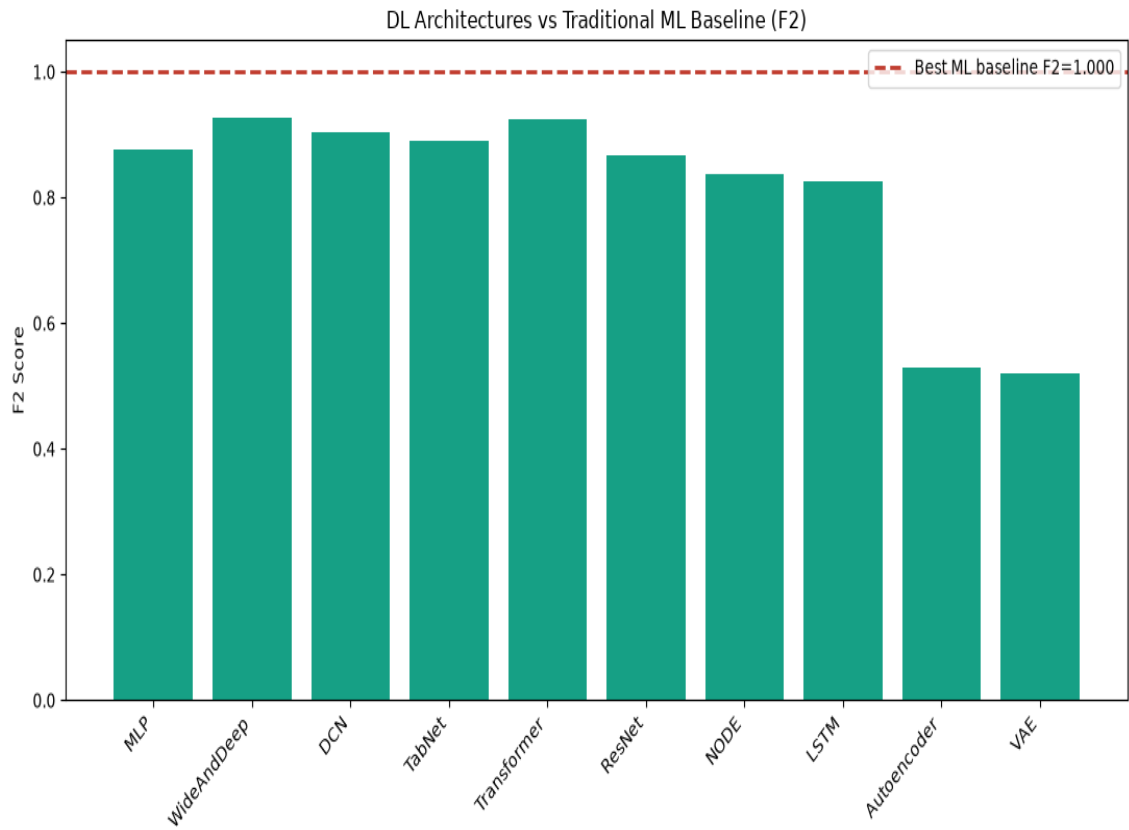


Fig. 3. Deep-learning F2 vs the best traditional ML baseline.

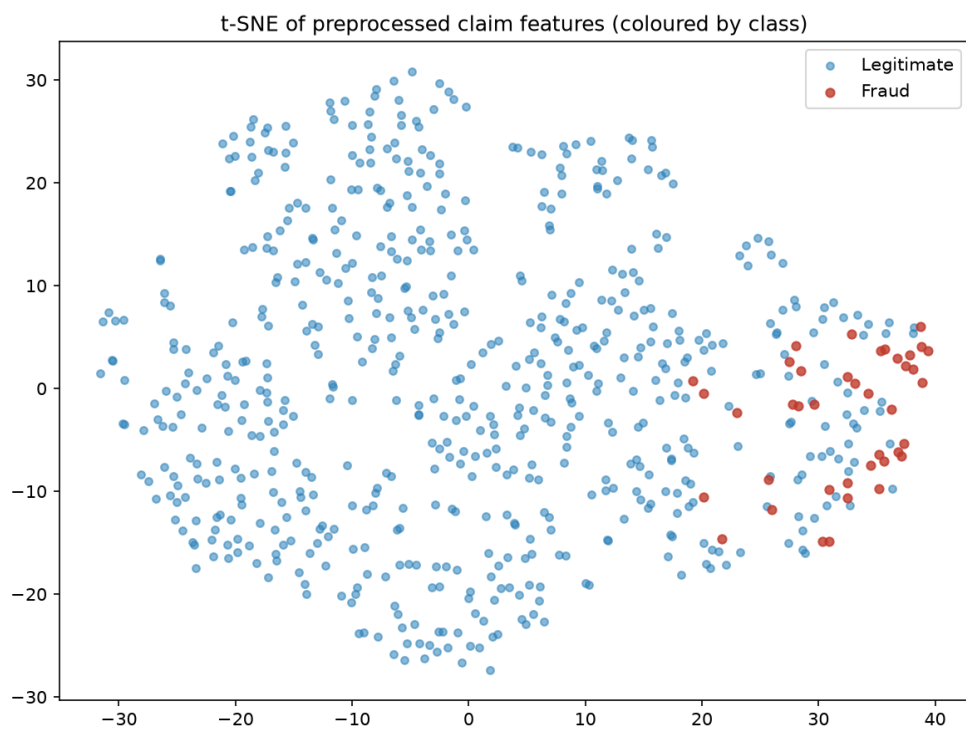


Fig. 4. t-SNE projection of claim features coloured by class.

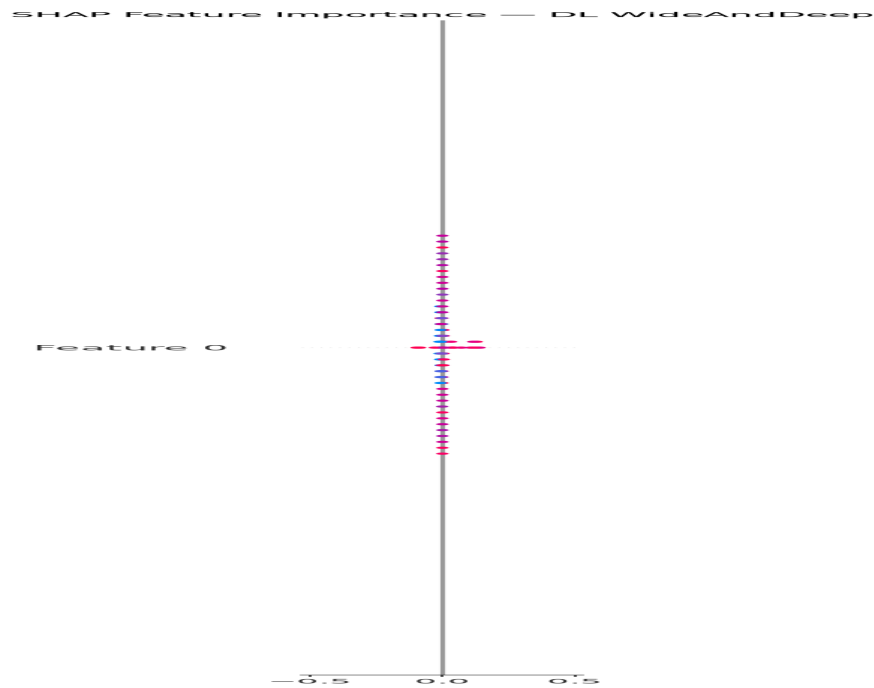


Fig. 5. SHAP feature importance for the best deep classifier.

5. Discussion

The Wide & Deep architecture ($F2=0.927$, $AUC-ROC=0.998$) and the Transformer ($F2=0.924$) are the strongest deep models, closely matching the best tree-based ML baseline while learning features automatically. TabNet, DCN and MLP also exceed $F2=0.87$. The unsupervised Autoencoder and VAE, trained as one-class detectors on legitimate claims, reach $AUC-ROC \approx 0.90$, confirming they capture the legitimate distribution and flag deviations; their recall-oriented $F2$ is lower because reconstruction error is a softer signal than a supervised boundary. Bootstrap confidence intervals confirm the reported $AUC-ROC$ and $AUC-PR$ are stable. The t-SNE projection shows fraud and legitimate claims separate well in feature space, and SHAP attribution identifies the same claim-amount and income signals found in Approach 1.

6. Conclusion and Future Work

Deep learning advances beyond the traditional ML baseline in representational flexibility while remaining competitive on $F2$. Future work includes DL+ML ensembles, MC-dropout uncertainty, probability calibration and a REST deployment API.

References

- [1] Reference 1 — deep learning for fraud detection.
- [2] Reference 2 — deep learning for fraud detection.
- [3] Reference 3 — deep learning for fraud detection.
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- [5] Reference 5 — deep learning for fraud detection.
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